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Effects of chronic stress on cognitive functions and anxiety related behaviors in rats

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The effects of 100 dB prenatal and chronic postnatal white noise stress (WNS) on some cognitive functions and behaviors in rats were investigated. For this purpose, 60 female Sprague-Dawley rats and their 90 male offspring were used. Pregnant rats were divided into Prenatal Stress (PS) and Prenatal Control (PC) groups. WNS was applied to PS group between the 14th and 21st days of their pregnancy, while PC rats were left undisturbed. After weaning, 40 male pups of PS dams were assigned to prenatal + chronic postnatal stress (PSCS) and prenatal stress + non-stress (PSN) groups. Pups of PC dams were divided into Control (CON) and Corticosterone (CORT) groups. During the postnatal 30th and 51st days, WNS was applied to PSCS and CORT rats everyday for 45 minutes, while PSN and CON groups were left undisturbed. The effects of stress on adult male offspring were investigated using Morris water maze and defensive withdrawal tests. Blood samples were collected after each test for serum corticosterone measurements. Blood samples of CORT rats were collected before the stress application and at the 1st, 7th, 14th, and 21st days of the stress period, immediately after cessation of the stress application. There were no significant differences among groups for learning and behavior tests. Corticosterone levels of CORT rats were significantly higher after the stress period than before stress application. These results indicate that although chronic 100 dB WNS induces a stress response by increasing corticosterone levels, it does not affect cognitive functions and anxiety related behaviors of adult male offspring.

Keywords: anxiety, learning, memory, rats, stress, white noise Corticosterone, Morris water maze test, defensive withdrawal tests, cognitive functions

Effects of stressful experiences on physiology and behaviors of animals and humans are quite complicated. Variation in intensity, duration, type of stressors and the age of the animal during stress exposure may produce important differences in patterns of stress responses (5, 14, 27, 30). Many studies investigated the effects of exposure to different stressors during the prenatal and postnatal periods of life. In these studies it was reported that stress increased or decreased or did not affect memory, learning, anxiety or fear related behaviors of rats (2, 7, 31, 35, 38, 39). For example, it has been demonstrated that prenatal unpredictable restraint stress induced spatial learning impairment in 1-month-old male rats but did not show any effect in 3-month-old male rats (40). On the other hand, has increased spatial learning performance been shown in male rats that were prenatally exposed to mild restraint stress (16). Furthermore, effects of chronic stress and the impact of stressful events on the adaptive capacities of adult subjects were investigated in some studies. However, obtained results are conflicting, both negative and positive effects of chronic stress in rats were reported and also persistency of the effects of stress varies in different studies (3, 20, 22, 28, 33). It has been

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reported that early-life chronic isolation stress increases anxiety-like and fear behaviors in adulthood (23). In another study, no alterations has been shown in anxiety-like behaviors in adult male rats that were prenatally exposed to restraint stress (6, 32). Taken together, results of these studies support the idea that adaptation to chronic stress, and its effects on behaviors, and persistency of effects of stressful events in later life depends on the stress regime and nature of the stressor (1, 13, 25).

It has been shown that postnatal environment and conditions modify the effects of prenatal stress (8, 16, 21, 38). For example, it has been suggested that anxious behavior developed as a consequence of a chronic prenatal restraint stress experience can be counteracted by early adoption (cross fostering) in rats (4). However, most of the studies investigating the influence of stress on rats have focused on the effects of either prenatal or postnatal stress. If we consider that environmental conditions are not changed, it is most likely that animals and people would still be exposed to the same stressor that they experienced during the prenatal period. In the present study, long-term effects of prenatal and chronic postnatal audiogenic stress were examined in adult male rats at the behavioral levels. The Morris water maze was used to measure spatial learning and memory performance and a defensive withdrawal test was used to assess anxiety like behaviors in rats. Serum corticosterone levels were measured to assess stress induced changes in HPA axis response.

Materials and Methods

Animals and housing

Sixty naive adult female (205±3 g body weight) and 12 naive adult male Sprague-Dawley rats (305±9 g body weight) obtained from the Institute for Experimental Medicine of Istanbul University were bred in our laboratory. They were housed with free access to food and water in a temperature controlled room (21±1 °C) under a 12:12 h light-dark cycle (lights on at 7:00 a.m.). Female rats were grouped (4–5 rats per cage) in clear plastic cages (46×32×14 cm) with wood chip bedding in the presence of a male for 5–6 days until pregnancy. Vaginal smears were examined every morning daily and the day when the smear was sperm positive was determined as gestational day 0. Thereafter, pregnant animals were housed individually.

All efforts were made to minimize the suffering and the number of animals used. The Ethic Committee of Istanbul University Veterinary Faculty approved the experimental protocol of the study.

Stress procedures

Pregnant rats were randomly divided into Prenatal Stress (PS, n=30) and Prenatal Control (PC, n=30) groups. PS rats were exposed to 45 min long white noise stress at the intensity of 100 dB (decibel-A scale) SPL (sound pressure level) from day 14 to 21 of pregnancy. During this period, stress was applied once a day at random times of the day. Stress application was performed in another room and animals were carried away and back during this period. The rat cages were arranged to ensure a similar distribution of noise to all the animals. Noise intensity was measured by placing a sound level meter (CEM DT–8820) in the cages at varied locations and calculating the mean value of different readings. Rats in the PC group were treated in the same way without audiogenic stimulation.

The offspring were weaned 21 days after birth and only litters consisting of five to ten pups with approximately half of them females and half of them males were kept for the study. To minimize litter effects, no more than 2 male offspring from each litter were assigned to each experimental group. Forty male offspring of the PS group were randomly divided into Prenatal Stress + Chronic Stress (PSCS, n=20) and Prenatal Stress + Non-Stress (PSN, n=20) groups. Fifty offspring of the PC group were randomly divided into Control (CON, n=20) and Corticosterone (CORT, n=30) groups. While the same stress procedure was applied to PSCS and CORT rats from day 30 to 51 of postnatal period, PSN and CON rats were not exposed to noise stress.

Corticosterone assay

Corticosterone levels were determined in blood samples of CORT (n=30) rats before stress application and at the 1st, 7th, 14th, 21st days of the stress period immediately after cessation of stress. Blood samples of 6 animals were collected at each sampling time point; sampled animals were not included in ongoing stress application process. Corticosterone levels of PSCS, PSN and CON rats were determined at the end of the behavioral tests.

Animals were anesthetized with xylazine (0.5 mg/100 g) and ketamine (4 mg/100 g). Blood samples were obtained by transcardially withdrawing 0.5 ml of blood into plain tubes. The blood was allowed to clot at room temperature. After that, it was centrifuged at 3500 rpm for 15 min and the serum was collected into eppendorf tubes and stored at –20 °C until assayed. The serum corticosterone level was determined using a commercial [^{25}I] corticosterone radioimmunoassay kit (MP Biomedicals), according to kit protocol. The sensitivity of corticosterone assay was 7.7 ng/ml. Corticosterone levels are expressed as ng/ml.

Behavioral testing

The testing period was started when the animals were young adults, at 70 days of age. Ten animals from each PSCS, PSN and CON groups were tested in the Morris water maze (MWM) and in the Defensive withdrawal test (DWT). All tests were conducted at the same time of the day beginning at 10:00 a.m. Experimental and control groups were tested simultaneously. The behavior of the rats was videotaped and then analyzed by experienced observers.

Morris water maze

The water maze was a black plexiglas circular pool (60 cm high, 132 cm diameter) that was filled with water (23±1 °C) to a level of 45 cm. A black escape platform (8 cm diameter) was used. The pool was divided by imaginary lines, into four quadrants and four different starting points were equally spaced around the perimeter of the pool. In the testing room, there were numerous constant visual cues outside the tank such as different shaped posters, a table, a door and the experimenter.

The rats were adapted to the experimental conditions a day before the testing period. They were launched into the water and allowed to swim freely for 1 min (no platform). For the spatial learning acquisition test, the escape platform was placed 2 cm below the surface in a fixed location in one of the four quadrants. It was equidistant from the sidewall and the middle of the pool. A block of four trials for each animal was performed per day, for 5 consecutive days. For each trial block, a rat was placed in the water facing the pool wall at one of the four randomly assigned starting points. Each starting point was used once a day and their sequence remained the same for all animals. The rat was allowed to swim for a maximum of 60 seconds to find the platform. Once the platform was located, the rat was allowed to remain there for 15 seconds. If an animal could not find the platform in this time, it was guided to the platform by the experimenter and allowed to stay there 15 seconds. The inter-trial interval for each rat was 30 seconds. Both the latency to find the platform and the amount of time spent in the quadrant where the platform was located were measured. One day after the acquisition period, the first probe trial was conducted. The platform was removed from the water and the rats were given one swim for 60 seconds. The time spent in the quadrant where the removed platform had been was recorded. The reversal test was carried out for three consecutive days beginning with the day after the first probe trial. The platform was hidden and placed in the opposite side from the place where it had been in the acquisition test. All the other experimental procedures were the same as during acquisition. The next day the second probe trial was performed. The cued test was carried out on the last day to evaluate the sensorimotor abilities of the animals. The escape platform was visible above the surface and had a coloured flag pole. Each animal had four trials starting from randomly selected starting points. The rat was required to locate the visible platform in 60 seconds. The latency to find the platform was recorded.

Results from blocks of 4 trials for each day of acquisition and reversal tests were averaged for every rat for the statistical analyses.

Defensive withdrawal test

The method of Ward et al (139) was used for the test. The test apparatus was a transparent plexiglas open arena (60×60×35 cm high) with black bottom and adjacent side compartment (15×20×15 cm high). Light intensity in the centre of the open field was 100 lx and was less than 1 lx in the dark box. The first day rats were introduced into the unfamiliar test environment by placing them into the open field and allowing them to explore freely for 15 min., although the animals' access to dark box was prevented. The following day, at the beginning of the test, the rats were placed in the dark box via the upper side lid and allowed free exploration in the open field. The test was conducted for ten minutes and was videotaped. After behavioral testing of each animal, the rats were returned to their home cage and the test apparatus was cleaned with 1% glacial acetic acid to prevent olfactory cues.

Latency to exit the dark box with all four paws for the first time, number of exits from the dark box, and time spent in the open field were determined.

Statistical analyses

Repeated measures analyses of variance (ANOVA) was applied to the data derived from MWM acquisition and reversal tests. Results of the probe and cued tests of MWM, defensive withdrawal tests and corticosterone levels were analyzed using one-way ANOVA. Differences were considered to be significant if $p<0.05$.

Results

Water maze performance

Spatial learning capacities are represented by the latencies to find the platform. Table I shows the mean latencies to find the platform across days for the three experimental groups. The repeated-measures ANOVA showed that rats in all groups progressively improved their ability to locate the platform over the days of acquisition ($F_{4, 104} = 32.38$, $p<0.001$) and reversal tests ($F_{2, 52} = 12.30$, $p<0.001$). While the effect of treatment was not significant, the interaction between group and days was not significant either. For the probe tests, there was no significant difference between the groups with respect to the percentage of time spent in the quadrant where the removed platform had been located (Fig. 1). On the cued test, there was no significant difference between groups, indicating that all groups had comparable levels of performance in the visible platform task (Fig. 2).

Table I. Latency to locate the platform in an acquisition and reversal tasks

Acquisition trial days	Escape latencies (sec)		
	PSCS	PSN	CON
1	47.06 ± 3.43	51.80 ± 3.61	39.55 ± 3.43
2	31.23 ± 4.44	37.18 ± 4.69	40.36 ± 4.44
3	21.86 ± 3.63	30.51 ± 3.83	23.60 ± 3.63
4	15.09 ± 4.41	26.94 ± 4.65	27.33 ± 4.41
5	16.94 ± 3.87	22.54 ± 4.08	16.22 ± 3.87
Reversal trial days			
1	23.80 ± 4.48	22.75 ± 4.72	16.89 ± 4.48
2	12.92 ± 2.65	14.43 ± 2.80	12.95 ± 2.65
3	10.93 ± 1.97	11.02 ± 2.08	10.93 ± 1.97

Data are the means ± SEM (sec) for the days of testing for PSCS, PSN and CON groups. Latency to find the platform decreases across acquisition trials ($F_{4, 104} = 32.38$, $p<0.001$) and reversal trials ($F_{2, 52} = 12.30$, $p<0.001$). The effect of treatment and group × trial interactions were not significant

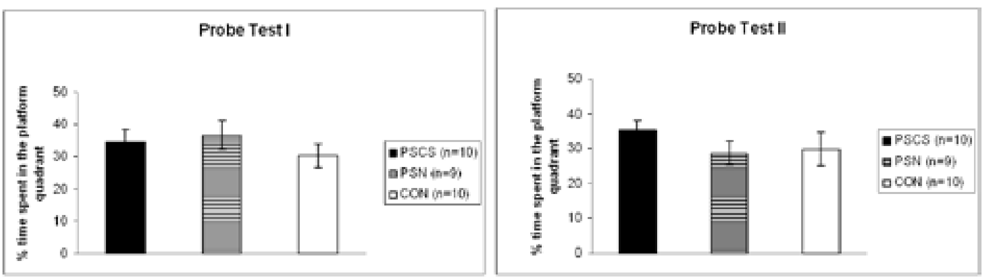

Fig. 1. Probe tests I and II. The percentage time (means ± SEM) spent in the quadrant where the platform had been located for PSCS, PSN and CON groups in the probe trials

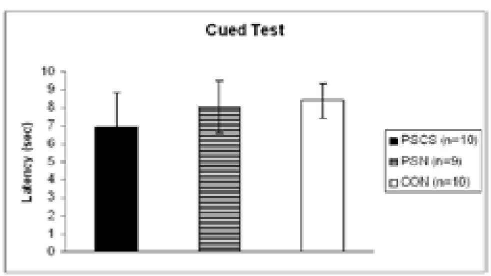

Fig. 2. Time (means ± SEM) taken to locate the platform for PSCS, PSN and CON groups in the cued trial

Defensive withdrawal test

There were no significant differences between groups in terms of measured parameters of defensive withdrawal test ($p>0.05$) (Table II). The variance between animals was high.

Table II. Parameters measured during the defensive withdrawal test

Parameters	Treatment groups		
	PSCS (n=9)	PSN (n=9)	CON (n=10)
Latency to exit the dark box (sec)	96.67 ± 63.80	137.00 ± 66.90	90.90 ± 58.69
Number of exits from the dark box	3.33 ± 1.21	3.67 ± 0.73	4.30 ± 0.82
Time spent in the open field (sec)	95.72 ± 55.32	66.67 ± 16.87	101.74 ± 25.25

Data are the means ± SEM. No significant differences were found between groups by one-way ANOVA

Corticosterone levels

Corticosterone levels of CORT group animals were significantly higher on days 1st, 7th, 14th and 21st of the stress period compared to the day before stress application and corticosterone levels of animals on day 14 were higher than on day 1 ($p<0.001$) (Table III). There was no significant difference in corticosterone levels between the experimental groups after behavioral tests (data not shown).

Table III. Corticosterone levels of CORT group animals (n=30)

	Before stress	After stress			
	day 1 (n=6)	day 1 (n=6)	day 7 (n=6)	day 14 (n=6)	day 21 (n=6)
Serum corticosterone levels (ng/ml)	148 ± 57*	303 ± 67	343 ± 31	491 ± 57**	365 ± 41

Data are the means ± SEM.
* Corticosterone levels were significantly higher on 1st, 7th, 14th, 21st days of the stress period than before stress application and **corticosterone levels on day 14 were higher than on day 1 after stress application ($p<0.001$)

Discussion

In the present work, effects of chronic white noise stress on spatial learning and memory, defensive withdrawal behaviors and corticosterone levels in rats were investigated.

Noise is one of the most widespread sources of environmental stress and white noise is an audiogenic stressor that produces stress response in the organism (11, 14, 17, 34). However, to our knowledge, there are only few studies investigating the effects of white noise stress on spatial learning and memory. Our results show that all animals were able to acquire and remember the spatial task, and that spatial learning and memory of animals in the Morris water maze were not affected by prenatal and postnatal chronic white noise stress treatment (Table I). It was reported that chronic (4 hours/day × 30 days) white noise (100 dB) exposure in adult rats significantly decreased the escape latency in the acquisition trial at days 1 and 3 also decreased the time spent in the target quadrant in the probe test in MWM (13). It was also suggested that chronic white noise stress (100 dB/4 hours per day/30 days) impaired spatial learning performance in radial arm maze in adult rats when measured an hour after noise stress exposure (25). Our results are contradictory to those two studies. They measured spatial abilities of animals immediately after the stress exposure. The lack of effects of white noise stress on spatial learning and memory in our study can be accounted for by evaluating the learning performance of animals at adulthood, not immediately after the cessation of stress period. In addition, stress regimes are different, i.e. the duration of stress application per day was shorter in our study. There are various studies that have investigated the effects of different types of stressors on learning and memory of rats and they have documented a variety of different results (2, 22, 28, 35). Discrepancies in results of the studies support the idea that effects of stress on cognition vary depending on the type and intensity of stress, application period and stress procedure, as well as on the time of determination of stress effects.

There were no statistically significant differences between the groups in the defensive withdrawal test (Table II). That is to say, our stress treatment did not show long-term effects on the behaviors of rats in the defensive withdrawal model of anxiety. Different types of stressors with varied intensities activate distinct neural pathways in the brain (14, 19, 29). It was reported that CRF and CRF receptors may be the mediators of anxiety (15, 36) and defensive withdrawal behaviors of rats are related with content and release of CRF in certain areas of the brain such as amygdala (37, 39, 41). It was determined that although CRF neurons in the hypothalamic paraventricular nucleus and medullary catecholaminergic cells participated in the organization of responses to short duration auditory stimulation (105 dB white noise), CRF neurons in the central nucleus of the amygdala were not activated by the stimulus (17). Similarly, it was reported that neuronal activation of the CRF containing neurons that contain CRF in the central nucleus of the amygdala was not observed as a result of 100 dB white noise exposure (14). Neuro-hormonal basis of formation of defensive withdrawal behaviors are quite complicated and our stress treatment might not have affected CRF containing neurons in the amygdala, which plays an important role in the formation of defensive withdrawal behaviors. Brain regions related to formation of defensive withdrawal behaviors may not be involved in stress responsiveness to auditory stimulation. However, involvement of the other neuroendocrine pathways such as central noradrenergic systems in the formation of defensive withdrawal behaviors (41) should be also considered. Further research is needed to investigate the relation between neuroendocrine responses to audiogenic stress and anxiety-like behaviors.

It has been suggested that white noise application with 90 dB and higher intensities evokes the plasma corticosterone levels (11). In the present study it was determined that corticosterone levels of animals were significantly higher on the 1st, 7th, 14th, 21st days of the stress period compared to the day before stress application (Table III.). Therefore, it may be suggested that chronic white noise stress with 100 dB intensity is sufficient to evoke stress responses of animals, i.e. the given noise was a definite stressor, and adaptation of HPA axis did not develop during the stress application period. This can be attributed the accumulative effect of repeated noise exposure and to the changes in the glucocorticoid negative feedback sensitivity in rats (26). However, results of the studies are conflicting, for example it was reported that rats exposed to chronic white noise (105 dB)/30 min per day for 15 days displayed strong habituation. Initially elevated corticosterone levels during the initial day of noise exposure significantly decreased with additional noise exposure in the chronically stressed group (10). On the other hand, it was shown that after white noise stress exposure, 100 dB/ 4 h per day for 30 days, plasma corticosterone levels significantly increased in chronically stressed rats (24, 33). Supporting our results, this sustained increase in corticosterone observed in the chronic group suggests that no adaptation of the HPA axis to the 30-days chronic noise exposure develops. Although inconsistent results have been reported in the literature regarding habituation to chronic stress, it seems that other factors are involved, such as the type of the stressor, intensity of the stressors and stress modalities (1, 13, 25). Following behavioral testing there were no significant differences between the treatment groups considering corticosterone levels. It was reported that corticosterone levels of stressed animals were higher during the stress period but returned to baseline 15 days after the stress period (7). These results are somewhat comparable to ours. The function of the HPA axis in stressed rats may not be permanently changed and may be completely recovered until the time of testing, as only some stressors are able to cause long-lasting changes in HPA responsiveness (13) and this may account for the lack of elevations in the corticosterone levels of our stressed animals after behavioral tests.

Conclusions

A growing body of literature demonstrates that effects of prenatal and postnatal stresses vary depending on many factors such as type, duration, severity of the stress regime, and age of the animals at the time of stress exposure (5, 13, 14, 21, 22, 25, 31). Moreover, behavioral, cognitive and hormonal responses of animals to stress are quite complex and do not have to always occur in concert with each other (11, 16, 20, 25). As a conclusion it is suggested that although chronic white noise (100 dB) stress causes an increase in corticosterone levels during the stress application, it does not affect spatial learning, memory and defensive withdrawal behaviors of adult rats. To understand the consequences of stress exposure and response mechanisms to stressors, further research is needed to investigate the effects of stressors on behaviors, cognition and physiology of organism.

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REFERENCES

1. Aguilera G: Corticotrophin Releasing Hormone, receptor regulation and the stress response. Trends Endocrinol. Metab. 9, 329–336 (1998)
2. Aleksandrov AA, Polyakova ON, Batuev AS: The effects of prenatal stress on learning in rats in a Morris maze. Neurosci. Behav. Physiol. 31, 71–74 (2001)
3. Araujo APN, DeLucia R, Scavone C, Planeta CS: Repeated predictable or unpredictable stress: effects on cocaine-induced locomotion and cyclic AMP-dependent protein kinase activity. Behav. Brain Res. 139, 75–81 (2003)
4. Barros VG, Rodríguez P, Martijena ID, Pérez A, Molina VA, Antonelli MC: Prenatal stress and early adoption effects on benzodiazepine receptors and anxiogenic behavior in the adult rat brain. Synapse 60, 609–618 (2006)
5. Bowman RE: Stress induced changes in spatial memory are sexually differentiated and vary across the lifespan. J. Neuroendocrinol. 17, 526–535 (2005)
6. Bowman RE, MacLusky NJ, Sarmiento Y, Frankfurt M, Gordon M, Luine VN: Sexually dimorphic effects of prenatal stress on cognition, hormonal responses, and central neurotransmitters. Endocrinology 145(8), 3778–3787 (2004)

7. Bowman RE, Zrull MC, Luine VN: Chronic restraint stress enhances radial arm maze performance in female rats. Brain Res. 904, 279–289 (2001)
8. Brabham T, Phelka A, Zimmer C, Nash A, Lopez JF, Vazquez DM: Effects of prenatal dexamethasone on spatial learning and response to stress is influenced by maternal factors. Am. J. Physiol. Regulatory Integrative Comp. Physiol. 279, R1899–1909 (2000)
9. Bullitt E: Expression of c-fos-like protein as a marker for neuronal activity following noxious stimulation in the rat. J. Comp. Neurol. 296, 517–530 (1990)
10. Campeau S, Dolan D, Akil H, Watson SJ: c-fos mRNA induction in acute and chronic audiogenic stress: possible role of the orbitofrontal cortex in habituation. Stress 5(2), 121–130 (2002)
11. Campeau S, Watson SJ: Neuroendocrine and behavioral responses and brain pattern of c-fos induction associated with audiogenic stress. J. Neuroendocrinol. 9, 577–588 (1997)
12. Cui B, Wu M, She X: Effects of chronic noise exposure on spatial learning and memory of rats in relation to neurotransmitters and NMDAR2B alteration in the hippocampus. J. Occup. Health. 51(2), 152–158 (2009)
13. Dal-Zotto S, Martí O, Armario A: Influence of single or repeated experience of rats with forced swimming on behavioural and physiological responses to the stressor. Behav. Brain Res. 114, 175–181 (2000)
14. Dayas CV, Buller KM, Crane JW, Xu Y, Day TA: Stressor categorization: acute physical and psychological stressors elicit distinctive recruitment patterns in the amygdala and in medullary noradrenergic cell groups. Eur. J. Neurosci. 14, 1143–1152 (2001)
15. Dunn AJ, Berridge CW: Physiological and behavioral responses to corticotropin-releasing factor administration: is CRF a mediator of anxiety or stress responses? Brain Res. Rev. 15, 71–100 (1990)
16. Fujioka T, Fujioka A, Tan N, Chowdhury GM, Mouri H, Sakata Y, Nakamura S: Mild prenatal stress enhances learning performance in the non-adopted rat offspring. Neuroscience 103, 301–307 (2001)
17. Helfferich F, Palkovits M: Acute audiogenic stress-induced activation of CRH neurons in the hypothalamic paraventricular nucleus and catecholaminergic neurons in the medulla oblongata. Brain Res. 975, 1–9 (2003)
18. Herdegen T, Leah JD: Inducible and constitutive transcription factors in the mammalian nervous system: control of gene expression by Jun, Fos and Krox, and CREB/ATF proteins. Brain Res. Brain Res. Rev. 28, 370–490 (1998)
19. Herman JP, Cullinan WE: Neurocircuitry of stress: central control of the hypothalamo-pituitary-adrenocortical axis. Trends Neurosci. 20, 78–84 (1997)
20. Knuth ED, Etgen AM: Long-term behavioral consequences of brief, repeated neonatal isolation. Brain Res. 1128, 139–147 (2007)
21. Lemaire V, Lamarque S, Le Moal M, Piazza PV, Abrous DN: Postnatal stimulation of the pups counteracts prenatal stress-induced deficits in hippocampal neurogenesis. Biol. Psychiatry 59, 786–792 (2006)
22. Luine V, Martinez C, Villegas M, Magarinos AM, McEwen BS: Restraint stress reversibly enhances spatial memory performance. Physiol. Behav. 59, 27–32 (1996)
23. Lukkes JL, Mokin MV, Scholl JL, Forster GL: Adult rats exposed to early-life social isolation exhibit increased anxiety and conditioned fear behavior, and altered hormonal stress responses. Horm. Behav. 55(1), 248–256 (2009)
24. Manikandan S, Padma MK, Srikumar R, Jeya Parthasarathy N, Muthuvel A, Sheela Devi R: Effects of chronic noise stress on spatial memory of rats in relation to neuronal dendritic alteration and free radical-imbalance in hippocampus and medial prefrontal cortex. Neurosci. Lett. 399(1–2), 17–22 (2006)
25. Marin MT, Cruz FC, Planeta CS: Chronic restraint or variable stresses differently affect the behavior, corticosterone secretion and body weight in rats. Physiol. Behav. 90, 29–35 (2007)
26. Meaney MJ, Diorio J, Francis D, Widdowson J, LaPlante P, Caldji C, Sharma S, Seckl JR, Plotsky PM: Early environmental regulation of forebrain glucocorticoid receptor gene expression: implications for adrenocortical responses to stress. Dev. Neurosci. 18(1–2), 49–72 (1996)
27. Mello AAF, Mello MF, Carpenter LL, Price LH: Update on stress and depression: the role of the hypothalamic-pituitary-adrenal (HPA) axis. Rev. Bras. Psiquiatr. 25, 231–238 (2003)
28. Nishimura J, Endo Y, Kimura F: A long-term stress exposure impairs maze learning performance in rats. Neurosci. Lett. 273, 125–128 (1999)
29. Pacak K, Palkovits M, Yadid G, Kvetnansky R, Kopin IJ, Goldstein DS: Heterogeneous neurochemical responses to different stressors: a test of Selye's doctrine of nonspecificity. Am. J. Physiol. 275, R1247–R1255 (1998)
30. Pijlman FT, Wolterink G, Van Ree JM: Physical and emotional stress have differential effects on preference for saccharine and open field behaviour in rats. Behav. Brain Res. 139, 131–138 (2003)
31. Pijlman FT, van Ree JM: Physical but not emotional stress induces a delay in behavioural coping responses in rats. Behav. Brain Res. 136, 365–373 (2002)

32. Richardson HN, Zorrilla EP, Mandyam CD, Rivier CL: Exposure to repetitive *versus* varied stress during prenatal development generates two distinct anxiogenic and neuroendocrine profiles in adulthood. Endocrinology 147(5), 2506–2517 (2006)
33. Roskoden T, Linke R, Schwegler H: Transient early postnatal corticosterone treatment of rats leads to accelerated aquisition of a spatial radial maze task and morphological changes in the septohippocampal region. Behav. Brain Res. 157, 45–53 (2005)
34. Samson J, Sheeladevi R, Ravindran R, Senthilvelan M: Stress response in rat brain after different durations of noise exposure. Neurosci. Res. 7(1), 143–147 (2007)
35. Sousa N, Lukoyanov NV, Madeira MD, Almeida OF, Paula-Barbosa MM: Reorganization of the morphology of hippocampal neurites and synapses after stress-induced damage correlates with behavioral improvement. Neuroscience 97, 253–266 (2000)
36. Takahashi LK, Ho SP, Livanov V, Graciani N, Arneric SP: Antagonism of CRF(2) receptors produces anxiolytic behavior in animal models of anxiety. Brain Res. 902, 135–142 (2001)
37. Takahashi LK, Kalin NH, Vanden Burgt JA, Sherman JE: Corticotropin-releasing factor modulates defensive-withdrawal and exploratory behavior in rats. Behav. Neurosci. 103, 648–654 (1989)
38. Vallee M, MacCari S, Dellu F, Simon H, Le Moal M, Mayo W: Long-term effects of prenatal stress and postnatal handling on age-related glucocorticoid secretion and cognitive performance: a longitudinal study in the rat. Eur. J. Neurosci. 11, 2906–2916 (1999)
39. Ward HE, Johnson EA, Salm AK, Birkle DL: Effects of prenatal stress on defensive withdrawal behavior and corticotrophin releasing factor systems in rat brain. Physiol. Behav. 70, 359–366 (2000)
40. Wu J, Song TB, Li YJ, He KS, Ge L, Wang LR: Prenatal restraint stress impairs learning and memory and hippocampal PKCbeta1 expression and translocation in offspring rats. Brain Res. 1141, 205–213 (2007)
41. Yang XM, Gorman AL, Dunn AJ: The involment of central noradrenergic systems and corticotropin-releasing factor in defensive withdrawal behavior in rats. J. Pharmacol. Exp. Ther. 255, 1067–1070 (1990)